

Foundation of Blockchain Technology (IS501)

Pre-Reading Material – Experiment 1

Setup of Remix IDE for Smart Contract Development

Estimated Reading Time: 20–25 minutes

Introduction

Blockchain technology has emerged as one of the most significant innovations in computing, enabling secure, transparent, and decentralized digital transactions. Unlike traditional systems that rely on a central authority, blockchain distributes data across multiple computers, making it more secure and resistant to tampering.

Today, blockchain technology is used in cryptocurrencies, banking, healthcare, supply chain management, digital identity, and many other applications.

The purpose of this experiment is to familiarize students with **Remix IDE**, an online development environment used for creating Ethereum smart contracts.

What is Blockchain?

A blockchain is a **distributed digital ledger** that records transactions in chronological order. These transactions are grouped into blocks, and each block is connected to the previous one using cryptographic techniques, forming a secure chain of information.

Characteristics of Blockchain

- Decentralized
- Transparent
- Secure
- Immutable
- Distributed

Because every participant has a copy of the ledger, altering recorded information becomes extremely difficult.

What is Ethereum?

Ethereum is a blockchain platform designed not only for recording transactions but also for executing computer programs known as **smart contracts**.

Unlike Bitcoin, which primarily supports digital currency transactions, Ethereum allows developers to build decentralized applications (DApps) for various real-world problems.

Examples include:

- Digital voting
- Supply chain tracking
- Healthcare record management
- Online auctions
- Banking and finance

What are Smart Contracts?

A smart contract is a program stored on a blockchain that automatically executes when predefined conditions are met.

Instead of depending on a third party, the blockchain executes the contract automatically, making the process transparent, reliable, and secure.

Some advantages include:

- Automatic execution
- Reduced human intervention
- Transparency
- Improved security
- Permanent record of transactions

What is Remix IDE?

Remix IDE is a **browser-based software development environment** used to create Ethereum smart contracts. It allows developers to write, compile, test, and deploy smart contracts directly from a web browser without installing additional software.

During this laboratory session, students will use Remix IDE to:

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Course Repository: <https://github.com/rajesh101191/Foundation-of-Blockchain-Technology>



- Create a Solidity file.
- Compile the program.
- Deploy a sample smart contract.
- Observe how a smart contract is executed in a virtual blockchain environment.

Why Do We Need Remix IDE?

Developing blockchain applications requires specialized software. Remix IDE simplifies this process by providing all the necessary tools in one place.

Some important features include:

- Browser-based development
- No installation required
- Built-in compiler
- Easy testing environment
- Beginner-friendly interface

Learning Outcome of this Experiment

After completing this experiment, students should be able to:

- Explain the purpose of Remix IDE.
- Open and navigate the Remix interface.
- Create a new Solidity project.
- Compile a simple smart contract.
- Deploy the contract using the JavaScript Virtual Machine (VM).

Before Coming to the Lab

Students should:

- Understand the basic concepts of blockchain.
- Know the purpose of Ethereum.
- Be familiar with the concept of smart contracts.
- Bring a laptop with a modern web browser (Chrome, Edge, or Brave) and an internet connection.

Self-Check Questions

1. What is blockchain technology?
2. Why is blockchain considered secure?
3. What is Ethereum?
4. What is a smart contract?
5. Why is Remix IDE used?
6. Name two applications of blockchain technology.

